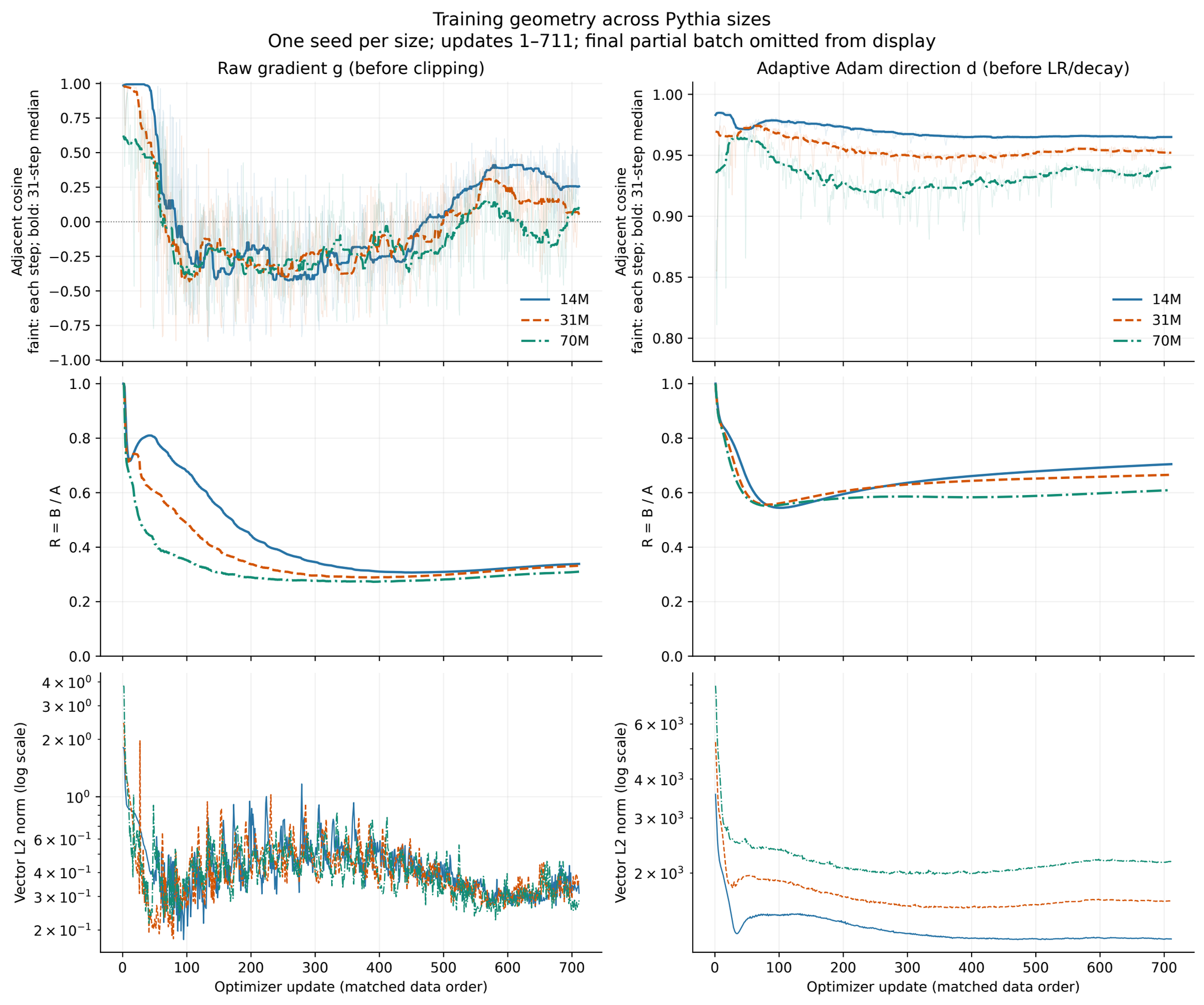

Training geometry across Pythia sizes
One seed per size; updates 1–711; final partial batch omitted from display
Raw gradient g (before clipping)
Adaptive Adam direction d (before LR/decay)
Adjacent cosine
faint: each step; bold: 31-step median
R = B / A
Vector L2 norm (log scale)
Optimizer update (matched data order)
14M
31M
70M